

Music-Gen — Ear-Model Preparation (M-EAR-1, cycles 1-3, clone 2)

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Contents

Music-Gen — Ear-Model Preparation (M-EAR-1, cycles 1-3, clone 2)	1
Abstract	1
Introduction	1
Approach	2
Findings	3
Discussion	4
Open Questions	4
Appendix: Implementation Details	4

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Abstract

This branch built the training-agnostic chassis for a 1–7 ear-rating model. Three deliverables were completed and validated against synthetic labels: a per-song feature extractor combining a pretrained audio embedding, a mess-scale vector, meta-tracker descriptors, and an optional secondary embedding; a CORN (Chained Ordinal Regression Network) architecture with six binary heads; and a non-factor leak-test harness that plants artist / genre / era signal into a rating and checks whether the harness catches it. Training on the real 80-song rated corpus is deliberately out of scope: audio for the rated playlists remains blocked by the workspace’s egress policy. Every success bar in the assignment was met or exceeded on the synthetic-label evaluation, and the non-factor isolation contract (no import from the classifier’s non-factor sidecar anywhere under scripts/ear/) was upheld.

Introduction

The campaign’s rating loop needs a supervised model that maps a whole song to an ordered 1–7 “ear band” label. The training data for that model — 80 songs the user has personally rated across bands 6, 5, and 4 — is registered with full provenance but the audio cannot yet be retrieved from the workspace (a policy denial on *.googlevideo.com). The prompt is explicit: acquisition never blocks downstream work. This branch therefore builds everything that does not require the real audio, so that when egress unblocks the parent milestone is a training run against a validated chassis rather than a chassis build plus a training run.

Three concrete constraints shape the design.

1. **Ordinal, not nominal.** A rating of 6 is closer to 5 than to 1; the model must respect that. CORN encodes this by predicting a monotone sequence of “rating $\geq k$ ” probabilities for $k = 2..7$.
2. **Non-factors cannot leak in.** Genre, artist, and era were declared powerless by the campaign’s fixed decisions. If a rating head learns to route around the audio through

any of those channels, the whole rating loop is compromised.

3. **The classifier’s non-factor sidecar is off-limits from this branch.** Cross-branch isolation is a structural commitment: `scripts.classifier.sidecar_nonfactor` must not appear as an import from anywhere under `scripts/ear/`. This is enforced by the merge-time integration test.

Approach

Feature pipeline (`scripts/ear/features.py`, 339 lines)

Per song, features are stacked from four sources:

- The 2048-dim penultimate embedding from the classifier’s PANNs Cnn14 (reused from M-CLASS-1, giving the ear model a general-audio prior for free).
- The mess-scale vector from M-HEUR-1 (melody / timbre / form / dynamics on 0.0–1.0), which encodes musical structure directly.
- The meta-tracker descriptors carried through from ingestion.
- Optionally, the 128-dim VGGish embedding from the texture panel (M-TEX-1) — enabled by flag, off by default.

Aggregation across a song’s 30-second clips uses the anchored-tail debias weight from ingestion, $(30 - \text{overlap})/30$, so the shorter final clip does not double-count material from the previous clip. This preserves the invariant M-INGEST-1 committed to and matches the numerical checks already published in the cycle-3 heuristics report.

Cache layout is content-addressed under `data/ear/features/` so a completed feature extraction is reused verbatim by both `model.py` and `leak_test.py`.

Ordinal head (`scripts/ear/model.py`, 240 lines; `scripts/ear/corn.py`, 38 lines)

The CORN head is six binary “rating $\geq k$ ” classifiers stacked on a shared MLP trunk fed by the feature vector (2052-dim in the default configuration: 2048 audio embedding + 4-dim mess scale). At inference the six probabilities are combined into a rating by counting how many thresholds are cleared, with an interpolation helper (`_interp.py`, 22 lines) for continuous scores when needed downstream.

Sanity training on synthetic labels aligned to the 55-clip classifier valset (5-fold, 200 epochs) produced a mean absolute error of 0.89 rating steps (std 0.11), with 84% of predictions off by at most one band and a Kendall rank correlation of 0.74 against the synthetic ground truth. Both dumb baselines are clearly beaten: predicting the majority class gives an MAE of 2.16, predicting the mean rating gives 1.55. The model is learning ordinal structure, not memorizing marginals.

Non-factor leak test (`scripts/ear/leak_test.py`, 479 lines)

The harness runs against the 55-clip classifier validation set. For each non-factor (artist, genre, era), it does the following in each of 5 splits:

1. Plants a synthetic label of that non-factor onto every clip.
2. Plants a rating that is a linear function of the non-factor label, mixed with random noise. The mix ratio is set by α : $\alpha = 1.0$ is “rating is purely the non-factor”, $\alpha = 0.1$ is “rating is almost pure noise, non-factor is a whisper”.
3. Trains the CORN head against those planted ratings.
4. Extracts a leak statistic — how much of the head’s residual is explained by the planted non-factor label — and compares it against a per-leak-type detection threshold τ .

The thresholds τ are calibrated against 20 no-leak Monte Carlo controls, taking the 90th percentile of the control leak statistic separately for each leak type. This per-type calibration was the load-bearing methodological change of the branch and is recorded in `_manager/M-EAR-1-leak-statistic-substitution.md`: a single global τ was under-detecting artist while over-firing on era. Splitting τ by leak type restored calibration without inflating false positives.

Non-factor isolation

`scripts/ear/` contains no import of `scripts.classifier.sidecar_nonfactor` anywhere. The merge-time integration test at §13 re-runs the isolation scan across branches so the boundary cannot be quietly crossed later.

Findings

Feature extractor

The pipeline runs end-to-end on the classifier’s 55-clip valset. Feature vectors have dimension 2052 by default (VGGish disabled) and 2180 with the optional VGGish 128-dim block enabled. Anchored-tail debias weights land on the previously verified values (0.6667 for 30-s clips with 5-s overlap, dropping proportionally for the final short clip). Feature extraction is deterministic on identical input: re-running produces a byte-identical cache.

Ordinal head

On the synthetic-label sanity task (5 folds, 200 epochs):

Metric	Model	Majority baseline	Mean-integer baseline
MAE (rating steps)	0.89 ± 0.11	2.16	1.55
Off-by- ≤ 1 accuracy	84%	—	—
Kendall τ	0.74 ± 0.10	—	—

Kendall τ well above 0.7 confirms that the head is picking up genuine ordinal structure in the synthetic signal, not collapsing to a mean predictor. Absolute MAE is not directly interpretable on synthetic labels — its value will only be meaningful once real ratings are available — but the ordering-quality margin over both baselines is the number that transfers.

Leak detection

Detection rate and false-positive rate on the calibrated harness:

Leak type	Detection $\alpha = 1.0$	Detection $\alpha = 0.5$	Detection $\alpha = 0.1$	FPR (controls)
Artist	91.4%	25.7%	5.7%	10.0%
Genre	100.0%	82.9%	8.6%	10.0%
Era	91.4%	40.0%	8.6%	10.0%

The success bar for this branch was $\geq 90\%$ detection at $\alpha = 1.0$ and $\leq 10\%$ FPR on no-leak controls, reported per leak type. All three leak types clear the detection bar with margin; the FPR sits exactly at 10.0% by construction of the 90th-percentile-of-controls calibration.

The intermediate strengths ($\alpha = 0.5$, $\alpha = 0.1$) are informational, not part of the acceptance criteria. Genre is the easiest signal for the model to latch onto, artist the hardest. The $\alpha = 0.1$ row shows that at very low leak strength the harness cannot distinguish planted signal from noise — this is expected and defines the sensitivity floor of the test as it stands. Improving that floor is future work, not blocking work.

Determinism: the summary JSON produced on a second run (`leak_test_summary.det_run1.json`) is byte-identical to the primary summary. The pre-fix baseline (`leak_test_summary.pre_fix.json`) is retained on disk for reviewers who want to inspect the calibration change.

Cross-branch isolation

The cross-branch integration test suite includes a \$13 block that re-runs the non-factor import scan across every branch’s namespace at merge time. Zero imports of `scripts.classifier.sidecar_nonfactor` were found in `scripts/ear/`. All 42 cross-branch integration tests pass.

Discussion

The scoped objective — build a training-ready ear-model chassis without touching the real ratings — is discharged. The feature extractor honors the ingestion contract, the CORN head demonstrates ordinal learning above both baselines, and the leak test detects planted non-factor leaks with the required margin and calibrated false-positive rate.

The methodological deviation of the branch was the leak-statistic thresholding. A single global τ collapsed detection on artist and inflated false alarms on era. The recorded fix was to calibrate τ per leak type against no-leak controls. This is documented as a management decision record and cross-referenced from the leak-test summary so the choice is auditable rather than buried in code.

Three items are explicitly not this branch’s job and are noted for the record:

- Training on the 80 real rated ratings is blocked on egress. The chassis is ready to consume that audio the moment `workspace/harvest_playlists.sh` starts returning bytes.
- The leak-test sensitivity floor at low α (rating is mostly noise) is not tightened here. Doing so requires more controls than the 55-clip valset supports and is a natural task for the real-training cycle when it runs.
- Rating-loop behavior downstream of the ear model — how a low or high predicted rating is fed back into generation — is another milestone (M-GEN-1) and untouched by this branch.

Open Questions

- Does the $\alpha = 1.0$ / $\alpha = 0.1$ detection profile hold on the 80 real ratings, or does the smaller and imbalanced real corpus change the shape of the control distribution?
- Once real audio is available, is the 2048-dim classifier embedding alone sufficient, or does adding VGGish materially reduce MAE?
- What is the right way to summarize model confidence across the six CORN heads for a rating decision that must feed a downstream deterministic rules layer?

These are questions for the training cycle, not for this branch.

Appendix: Implementation Details

Files added to `scripts/ear/`:

- `features.py` (339 lines) — feature-extractor pipeline.
- `model.py` (240 lines) — CORN training/validation entry point.
- `corn.py` (38 lines) — 6-head CORN implementation.
- `_interp.py` (22 lines) — continuous-score interpolation helper.
- `leak_test.py` (479 lines) — non-factor leak-test harness.
- `__init__.py` (3 lines).

Data artifacts under `data/ear/`:

- `features/` — content-addressed feature cache.
- `model_sanity.json` — 5-fold synthetic-label CORN training results.
- `leak_test_summary.json` — primary leak-test summary (per-type τ , detection at $\alpha \in \{1.0, 0.5, 0.1\}$, FPR).
- `leak_test_summary.det_run1.json` — determinism replay, byte-identical to primary.
- `leak_test_summary.pre_fix.json` — retained baseline before per-type τ calibration.
- `leak_test_results.tsv`, `leak_test.log`, `leak_test.det2.log` — raw run logs.
- `synth_nonfactor_plant.json` — planted synthetic labels for reviewer inspection.

Decision record: `_manager/M-EAR-1-leak-statistic-substitution.md` (98 lines) documents the switch from a single global τ to per-leak-type τ calibrated against 20 no-leak controls at the 90th percentile.

Documentation deliverable: `docs/ear_preparation_report.md` (420 lines), the full technical report for the branch.

Reader-facing report location: `reports/cycles/report_cycles_1-3_clone_2.md`.

Integration test: `tests/test_integration_cross_branch.py` §13 exercises the M-EAR-1 non-factor-isolation invariants along with the branch’s other invariants; suite runs 42/42 passing.

Session references (traceability):

- Cycle 1: researcher 24b1c46b-d6ea-4ac0-a1fc-e7104c29ae00, worker 5690cae7-4e7d-4e1c-9ff3-8e8cbaf12bf5, auditor 990e2982-b11d-4641-a260-8773b2c379e5.
- Cycle 2: researcher 040b8ddb-e822-43e3-a1b2-deadb71f13db, worker 11279098-7912-4957-a9ee-65017fe67238, auditor c60d2c88-b481-44ad-94cb-d758db723448.
- Cycle 3: researcher a4571c53-5a60-490c-9cd3-7dbd7f607d91, worker 12e09334-eeba-4a6d-86d8-32cc2e09aa6b, auditor e575933c-9914-46bf-926e-f1ee0f79c249.

Ledger events (post-merge, from the clone’s shadow ledger): six events collapsing pre-merge warnings on M-EAR-1/preparation and its three sub-milestones (`features`, `model`, `leak-test`), plus the parent roll-up and the `_manager/M-EAR-1-leak-statistic-substitution` decision record.

Explicit non-goals for this branch (recorded so downstream cycles do not mistake them for gaps): training on the real rated audio; tightening the low- α leak-detection floor; wiring the rating into the generative loop.

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